

## KAMLOOPS, BC, Canada

WMO: 718870

Lat: 50.6993N

Lon: 120.4406W

Elev: 1133

StdP: 14.10

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -0.6       | 5.7 | -11.8                         | 3.0 | 2.1  | -4.6 | 4.5 | 7.3  | 27.7                  | 28.0 | 24.2 | 28.0 | 7.4                   | 90   | 0.557 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 25.2                   | 93.5            | 64.3 | 89.8 | 63.3 | 86.0 | 62.0 | 66.0                | 88.1 | 64.5 | 85.3 | 63.1 | 82.5 | 6.8                  | 110  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 58.9                            | 77.5 | 69.8 | 57.0 | 72.4 | 68.9 | 55.3 | 67.9 | 68.6 | 31.3          | 88.3 | 30.1 | 85.6 | 29.1 | 82.9 | 72.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |       |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|-------|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |       |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |       |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |       |
| 22.8              | 20.1 | 18.0 | DB                         | -4.7 | 99.5               | 7.4 | 3.2                                                | -10.0 | 101.8      | -14.3 | 103.6      | -18.5 | 105.4      | -23.9 | 107.7 |
|                   |      |      | WB                         | -5.6 | 69.4               | 7.1 | 1.5                                                | -10.7 | 70.5       | -14.9 | 71.4       | -18.9 | 72.3       | -24.1 | 73.4  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 49.2        | 27.9  | 31.3  | 41.1  | 49.7  | 58.8  | 65.0  | 71.8  | 70.5  | 60.8  | 47.5  | 36.4  | 28.3  |
|                                                                        | DBStd       | 17.25       | 10.51 | 9.17  | 7.69  | 5.88  | 6.44  | 5.87  | 6.06  | 5.69  | 6.08  | 6.47  | 9.19  | 9.86  |
|                                                                        | HDD50       | 2777        | 685   | 523   | 282   | 74    | 7     | 0     | 0     | 0     | 2     | 123   | 410   | 671   |
|                                                                        | HDD65       | 6295        | 1150  | 943   | 739   | 458   | 212   | 71    | 15    | 18    | 155   | 541   | 857   | 1136  |
|                                                                        | CDD50       | 2485        | 0     | 0     | 8     | 66    | 278   | 449   | 675   | 634   | 325   | 47    | 3     | 0     |
|                                                                        | CDD65       | 527         | 0     | 0     | 0     | 0     | 18    | 69    | 225   | 187   | 28    | 0     | 0     | 0     |
|                                                                        | CDH74       | 6073        | 0     | 0     | 0     | 22    | 331   | 836   | 2534  | 2012  | 336   | 2     | 0     | 0     |
|                                                                        | CDH80       | 2509        | 0     | 0     | 0     | 2     | 86    | 308   | 1171  | 852   | 90    | 0     | 0     | 0     |
|                                                                        | WSAvg       | 6.9         | 7.1   | 6.7   | 7.9   | 7.5   | 6.8   | 6.5   | 6.1   | 5.7   | 5.8   | 7.0   | 7.8   | 7.6   |
|                                                                        | PrecAvg     | 11.3        | 1.1   | 0.6   | 0.4   | 0.6   | 0.9   | 1.3   | 1.2   | 1.2   | 1.1   | 0.7   | 1.0   | 1.3   |
| (15) Precipitation                                                     | PrecMax     | 15.4        | 2.8   | 1.7   | 1.0   | 2.8   | 2.2   | 2.9   | 2.8   | 4.9   | 3.4   | 2.1   | 2.7   | 4.3   |
|                                                                        | PrecMin     | 6.5         | 0.2   | 0.1   | 0.0   | 0.1   | 0.2   | 0.2   | 0.1   | 0.2   | 0.0   | 0.1   | 0.2   | 0.2   |
|                                                                        | PrecStd     | 2.0         | 0.6   | 0.4   | 0.2   | 0.5   | 0.5   | 0.7   | 0.7   | 0.9   | 0.7   | 0.4   | 0.6   | 0.7   |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 48.8  | 53.7  | 65.4  | 77.3  | 87.6  | 94.2  | 99.2  | 96.7  | 88.1  | 70.1  | 58.6  | 51.1  |
|                                                                        |             | MCWB        | 41.1  | 43.5  | 49.7  | 55.6  | 60.6  | 63.8  | 66.1  | 64.9  | 62.7  | 56.0  | 48.0  | 42.9  |
|                                                                        | 2%          | DB          | 44.2  | 48.7  | 59.7  | 70.2  | 81.9  | 87.9  | 94.8  | 92.4  | 82.2  | 64.6  | 53.7  | 45.7  |
|                                                                        |             | MCWB        | 38.8  | 40.9  | 46.2  | 52.2  | 58.3  | 61.5  | 64.8  | 64.2  | 61.3  | 52.8  | 46.0  | 39.8  |
|                                                                        | 5%          | DB          | 41.3  | 45.3  | 56.2  | 66.0  | 77.4  | 82.8  | 90.9  | 88.7  | 77.5  | 61.2  | 50.5  | 42.4  |
|                                                                        |             | MCWB        | 37.4  | 38.9  | 44.3  | 49.8  | 56.4  | 59.7  | 63.8  | 63.3  | 59.4  | 50.7  | 43.8  | 37.5  |
|                                                                        | 10%         | DB          | 38.8  | 42.3  | 52.8  | 62.0  | 72.8  | 78.4  | 86.8  | 84.8  | 73.3  | 57.8  | 47.6  | 39.3  |
|                                                                        |             | MCWB        | 35.7  | 37.4  | 42.6  | 47.7  | 54.7  | 58.4  | 62.4  | 62.1  | 57.4  | 49.1  | 42.5  | 35.6  |
|                                                                        | 0.4%        | WB          | 41.9  | 44.9  | 49.9  | 56.1  | 61.9  | 65.4  | 68.3  | 68.0  | 64.6  | 57.6  | 49.8  | 43.7  |
|                                                                        |             | MCDB        | 47.6  | 52.1  | 63.8  | 73.4  | 82.5  | 88.0  | 93.6  | 89.8  | 81.5  | 67.2  | 56.2  | 49.6  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB          | 39.3  | 41.8  | 47.3  | 53.5  | 59.4  | 62.9  | 66.4  | 65.9  | 62.4  | 54.5  | 46.9  | 40.5  |
|                                                                        |             | MCDB        | 43.3  | 47.1  | 58.0  | 68.3  | 77.8  | 83.9  | 89.5  | 87.4  | 78.7  | 62.7  | 52.0  | 45.0  |
|                                                                        | 5%          | WB          | 37.6  | 39.7  | 45.3  | 51.0  | 57.6  | 61.1  | 64.8  | 64.4  | 60.5  | 51.9  | 44.8  | 38.0  |
|                                                                        |             | MCDB        | 40.9  | 44.5  | 54.7  | 63.5  | 74.0  | 79.0  | 86.3  | 84.8  | 75.6  | 59.3  | 49.5  | 41.8  |
|                                                                        | 10%         | WB          | 36.0  | 37.7  | 43.2  | 48.8  | 55.8  | 59.4  | 63.4  | 62.9  | 58.5  | 49.7  | 42.9  | 35.8  |
|                                                                        |             | MCDB        | 38.5  | 41.9  | 51.6  | 60.3  | 70.4  | 75.2  | 83.5  | 81.6  | 70.9  | 56.5  | 47.1  | 39.0  |
|                                                                        | 5% DB       | MDBR        | 9.3   | 13.1  | 18.1  | 21.8  | 22.9  | 22.5  | 25.2  | 25.4  | 22.7  | 16.6  | 10.9  | 9.2   |
|                                                                        |             | MCDBR       | 11.2  | 15.5  | 23.1  | 28.6  | 31.7  | 30.8  | 32.6  | 32.7  | 30.1  | 20.9  | 13.6  | 12.0  |
|                                                                        |             | MCWBR       | 7.8   | 9.6   | 12.8  | 14.3  | 13.4  | 11.4  | 11.2  | 12.4  | 13.6  | 11.6  | 8.6   | 7.9   |
|                                                                        |             | MCWBR       | 10.5  | 14.5  | 21.9  | 26.0  | 28.0  | 27.1  | 29.4  | 29.9  | 27.7  | 19.1  | 12.4  | 11.3  |
| (35) Mean Daily Temperature Range                                      | 5% WB       | MCWBR       | 7.4   | 9.3   | 12.6  | 13.3  | 12.2  | 10.7  | 10.9  | 11.8  | 13.0  | 11.1  | 8.3   | 7.9   |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |
|                                                                        |             |             |       |       |       |       |       |       |       |       |       |       |       |       |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.302 | 0.298 | 0.334 | 0.370 | 0.393 | 0.388 | 0.370 | 0.383 | 0.354 | 0.338 | 0.313 | 0.291 |
|                                                                        |             | taud        | 2.304 | 2.406 | 2.382 | 2.313 | 2.291 | 2.340 | 2.418 | 2.385 | 2.447 | 2.458 | 2.413 | 2.349 |
|                                                                        | Ebn at Noon |             | 228   | 267   | 276   | 276   | 274   | 276   | 279   | 269   | 266   | 248   | 225   | 215   |
|                                                                        |             | Edh at Noon | 21    | 25    | 31    | 37    | 40    | 38    | 35    | 34    | 29    | 24    | 19    | 18    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 307   | 613   | 1007  | 1440  | 1726  | 1869  | 2037  | 1734  | 1210  | 701   | 333   | 234   |
|                                                                        | RadStd      |             | 28    | 48    | 68    | 119   | 137   | 162   | 158   | 100   | 101   | 71    | 36    | 24    |

## Historical Trends

|                              |  | DBAvg | Heating |        |       | Cooling |       |     | Degree-Days |       |       |       |
|------------------------------|--|-------|---------|--------|-------|---------|-------|-----|-------------|-------|-------|-------|
|                              |  |       | 99% DB  | 99% DP | 1% DB | 1% WB   | 1% DP |     | HDD50       | HDD65 | CDD50 | CDD65 |
|                              |  | (d)   | (e)     | (f)    | (g)   | (h)     | (i)   | (j) | (k)         | (l)   | (m)   |       |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |
| (47) Regional (21 neighbors) |  | N/A   | N/A     | N/A    | N/A   | N/A     | N/A   | N/A | N/A         | N/A   | N/A   |       |

Nomenclature: See separate page